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Group B Streptococcus maternal colonization and neonatal sepsis in Belgium between 2012 and 2021: a description of the epidemiological situation and identification of risk factors

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Abstract

Background Group B Streptococcus (GBS) infection is a leading cause of neonatal morbidity. Maternal rectovaginal GBS colonization is a primary risk factor for early onset neonatal GBS infection. In Belgium, pregnant women are screened and, if positive or at risk of transmission, offered intrapartum antibiotic prophylaxis (IAP). We aimed to provide the first comprehensive overview of Belgian data and identify possible risk factors of maternal GBS colonization and neonatal infection.

Methods We calculated proportions of maternal screening, colonization and incidence of all neonatal GBS infections and identified their risk factors using log binomial regression from national registries' data between 2012–2021.

Results Of all women, 90.2% were screened and of them, 18.5% were GBS colonized. These proportions were stable over time. Risk factors for maternal GBS colonization included Sub-Saharan (aRR = 1.45, [1.41–1.48]) and North-African origin (aRR = 1.31, [1.28–1.34]) and 5 + parity (aRR = 1.21, [1.18–1.25]).

Neonatal GBS sepsis incidence showed a gradually decreasing trend in two individual data sources. Flanders' overall neonatal GBS sepsis estimates were around 0.49 cases per 1000 livebirths. Risk factors for neonatal GBS infection included preterm birth (aRR = 7–41, depending on level of prematurity) and very low birthweight (aRR = 14.83, [6.48–33.94]). Babies of colonized women were at higher risk without (aRR = 7.05, [5.05–9.85]) than with IAP (aRR = 4.34, [3.28–5.74]) compared to non-colonized women.

Conclusions The maternal colonization rate and the neonatal GBS sepsis incidence in Belgium is comparable to European data. We identified already known risk factors and effectiveness of IAP. Mothers at risk for GBS colonization and premature/low birth weight deliveries could benefit most from additional maternal vaccination strategy.

Keywords Group B streptococcus (GBS), Maternal GBS colonization, Neonatal GBS infection, Pregnancy, Neonatal sepsis, Risk factors, Belgium

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Background

Invasive infection with *Streptococcus agalactiae* (group B *Streptococcus*, GBS) remains one of the leading causes of neonatal morbidity and mortality globally [1]. Invasive GBS neonatal disease (iGBS) is divided into early onset disease (EOD), when pneumonia, sepsis or meningitis develop in the first six days of life [1–3], or late onset (LOD, in the 7–89 days of life) with sepsis or meningitis. Complications of meningitis can lead to long term neurologic sequelae such as sight or hearing loss and cerebral palsy. Worldwide it is estimated that 231,000 cases of EOD and 162,000 cases of LOD occur yearly [4].

Maternal rectovaginal GBS colonization is a primary risk factor for EOD. In the absence of intrapartum antibiotic prophylaxis (IAP), about 50% of GBS colonized women transmit colonization to their newborn and 1–2% of those colonized develop EOD [3]. IAP can be used to prevent EOD, but there is currently no consensus on how to identify those women that would benefit from IAP. In the USA and many high-income countries, including Belgium, all women are screened for GBS colonization antenatally (universal screening) and receive IAP if positive [5–8]. In contrast, a risk-based approach, with IAP administered only in case of clinical risk factors, is preferred in the UK and the Netherlands, among other countries [2, 9–12]. Although universal screening has undoubtedly led to a decrease of EOD cases [13], GBS colonization can be permanent, intermittent or transient [14], meaning that universal screening can lead to antibiotic overuse. Large-scale use of antibiotics might contribute to antimicrobial resistance and might have a negative effect on the neonatal microbiome [15]. Additionally, IAP policies, especially universal screening, are resource-intensive and hence mostly limited to high-income countries [16]. Moreover, IAP is not effective in preventing GBS-associated stillbirth, preterm birth or LOD. Indeed, LOD is thought to be caused by close contact with the colonized mother mostly after birth or with other colonized family members after delivery, but can also be acquired nosocomially or from individuals in the community [17].

A maternal vaccine could be an overarching solution (regardless of onset time or country setting) and would also limit antibiotic use. Phase 2 studies with multivalent investigational maternal GBS vaccines are ongoing and the World Health Organization (WHO) recommends that by 2026, at least one affordable vaccine should be licensed and WHO-prequalified for maternal immunization during pregnancy [18, 19].

In Belgium, a guideline for *Prevention of Perinatal GBS Infections* exists since 2003 [20]. It recommends universal screening of rectovaginal GBS colonization

between weeks 35–37 of gestation. Administration of IAP (penicillin or ampicillin) is recommended to mothers positive on screening, with GBS bacteriuria, with a previous child with iGBS, or in case of preterm delivery < 37 weeks [14]. If no test result is available at time of delivery, IAP should only be administered in case of risk factors for GBS colonization such as intrapartum fever or membrane rupture ≥ 18 h [20]. Although the guideline has existed in unchanged form for 20 years, there has to date not been any formal evaluation of its implementation or impact. Therefore, we aimed to collect and analyze all available information to describe the past and present epidemiological situation of GBS maternal colonization and invasive neonatal infection in Belgium. Additionally, we explored possible predictors for colonization and invasive infection, as these could help to direct preventive strategies to those that would benefit most.

Methods

Data sources

We retrospectively collected observational data from existing registries. As there is a delay in availability of data, the study period was set from 2012 to 2021. Information on both maternal GBS colonization and neonatal invasive infection was available from the Minimal Hospital Data (MHD) and the birth certificates in the Belgian Birth Register (BBR). The MHD are anonymized administrative, medical and nursing data on all hospitalizations, which all non-psychiatric Belgian hospitals have to register. Diseases are coded using the *International Classification of Diseases* (ICD-10 from 2015, ICD-9 before) [21]. Due to the low data quality of ICD-9 coding (2012–2014), the two periods were not comparable, and 2012–2014 data had to be discarded. The BBR contains birth certificates for all live births (regardless of gestational age or birth weight), and stillbirths from 22 weeks or 500 g onwards. Regional databases from the Perinatal Epidemiology Center (CEpiP, Brussels-Capital Region and the Walloon Region) and the Study Center for Perinatal Epidemiology (SPE, Flemish Region) were merged. These include socio-economic data which are completed by the civil registration service or by parents when registering the birth/death within 15 days of delivery. The hospital medical data are collected and completed by care providers. The SPE additionally collects medical information on hospitalization linked to the BBR if a newborn is admitted to a neonatal ward during the first 7 days of life. Additional data on neonatal GBS infection were available from the National Reference Center for Group B *Streptococcus* (NRC), but coverage is not exhaustive as invasive strains are sent to the NRC on a voluntary basis.

Case definitions

A maternal GBS colonization case was identified if the corresponding box was ticked (BBR, NRC) or corresponding ICD-10 code registered (MHD), see Supplementary Material for more details. Belgian guidelines suggest culture-based diagnosis for prenatal maternal GBS screening at week 35–37 with the recommendation to use a selective enriched broth medium further subcultured onto a GBS-specific medium when growing GBS.

An invasive neonatal GBS infection was identified if it was recorded in BBR, MHD or NRC. EOD is defined as GBS neonatal infection starting in the first six days of life, LOD in 7–89 days. The details of the case definitions and ICD-10 codes registered (MHD) can be found in the supplement. For the risk factor analysis using BBR data from Flanders, we included as ‘neonatal GBS infections’ all babies that had a proven GBS infection AND were admitted to a neonatal ward in the first 7 days of life

- a) because of proven infection in the <7 days of life.
- OR
- b) for any reason other than congenital malformations AND developed a severe infection during their stay in the neonatal ward.

Exact time of onset of infection was not available in this dataset, so that we could not formally distinguish EOD and LOD. In NRC data, clinical manifestations like respiratory tract infection and meningitis could be reported separately, but a code for sepsis was lacking. Therefore, we used confirmed invasive infection without focus as a proxy for sepsis.

Statistical analysis

We described the demographic and clinical characteristics of the maternal study population. Using BBR and MHD data, we calculated proportions of maternal screening and colonization, as well as case numbers and reported incidence of neonatal invasive infection over time and by gender. To compare trends and validate magnitude, we compared neonatal GBS sepsis to overall bacterial sepsis incidence over time. Using MHD, BBR and NRC data, demographic and clinical characteristics of neonates with GBS, proportion of antibiotic-resistant strains, and incidence and notification rate of neonatal infection were described. We calculated means for continuous variables and proportions for categorical variables, as well as annual incidence per 1000 livebirths and its evolution over time.

To explore possible predictors for maternal GBS colonization and neonatal GBS infection, we conducted univariable and multivariable analyses using log binomial

models in a complete case analysis (missing data were assumed to be completely at random). Potential risk factors identified in literature were included in the multivariable model. (Adjusted) risk ratios ((a)RR) and their 95% confidence intervals (CI) were calculated to provide insights into the association between risk factors and the outcome.

For maternal GBS colonization, risk factors are biological, such as maternal age [22, 23], increased number of deliveries [22, 24], and obesity [24], as well as socioeconomic, including ethnicity [24] and illiteracy [25]. The BBR did not contain information on ethnicity or illiteracy, but we used socioeconomic data on maternal nationality at delivery, educational level and work status as a proxy. We also included data on maternal hypertension and diabetes. *P*-values were obtained using likelihood ratio tests. Additionally, for variables with 3 or more categories, the *p*-value of an omnibus test was reported. To even out the possible effect of different practices in hospitals (e.g., different sensitivity of lab tests and different screening practices), we used a random intercept for clustering by hospital.

A similar approach was used to model overall neonatal GBS infection. GBS infected babies were compared to healthy neonates. We excluded babies with congenital malformations and those admitted to neonatal wards for reasons other than infection. Risk factors for EOD previously identified in literature are maternal colonization [26, 27], preterm birth [26], low birth weight, prolonged rupture of membranes [28], intrapartum fever [28], chorioamnionitis [28], maternal demographic factors (e.g., age and ethnicity) [1, 29]. We included all respective available BBR variables: gender and birthweight of the baby, maternal colonization combined with IAP administration, maternal age, gestational age and maternal nationality as potential risk factors. The number of levels when categorizing variables and the number of variables included were limited due to the low neonatal GBS case numbers in the study population.

All analyses were performed in R (version 4.3.2 using *glm* and *glmer* packages) and MS Excel.

Results

BBR data was available for 1,225,761 deliveries in Belgium between 2012 and 2021. After exclusion of 7,502 records from stillbirths and deduplicating 20,265 records from pregnancies of multiples, our study population comprised 1,197,994 women who gave live birth. BBR data were near-complete (missing $\leq 1\%$ of outcome cases) for all variables except BMI, maternal education and maternal work status (respectively, 5.0%, 14% and 6% missing).

The mean maternal age was 30.6 years, women had an average of 1.9 pregnancies (including current delivery), mean length of gestation was 38.8 weeks and the average BMI before pregnancy was 24.9 kg/m². Of all included women, 46% had received tertiary education, 69% had a paid employment and 77% had Belgian nationality at delivery.

Maternal GBS colonization

In the BBR dataset, overall, 115,744 (9.7%) pregnant women were not screened for GBS and 199,669 (16.7%) were colonized at the time of screening. The proportion of both GBS colonized (range 15.4%–17.4%) and non-screened pregnant women (range 8.9–11.2%) remained stable during the study period (Table 1). Of non-screened women, 32.5% had a planned caesarean section (CS) and 17% delivered before 35 weeks. When discarding those births, the proportion of non-screened dropped to 6%. Although this proportion was generally stable over the study period, there was a peak of 7.3% during the 2020 pandemic year. Of all women with a known GBS screening result, 18.5% were reported to be GBS colonized, and this proportion remained stable over time (range 17.1%–19.3%). Of the colonized women 85% received IAP.

The province of Antwerp and the Brussels-Capital Region showed a markedly higher maternal colonization rate (18.5% and 18.4%, respectively) than the other provinces (range 14.9%–16.7%). The lowest rate was found in Walloon Brabant (14.9%) (Supplementary Table S1).

In the MHD dataset, maternal GBS colonization rate was 15% (range 14.1%–15.7%) during the study period, slightly lower than the BBR estimates, but with the same stable trend over time (data not shown). The

MHD data also confirmed the variation between provinces, with the highest proportions in Brussels and Antwerp (16.8% and 15.7%, respectively).

Table 2 presents the results of the maternal GBS colonization risk factor analysis. Due to the large size of the dataset, all included variables showed a statistically significant yet relatively weak association with the outcome. In multivariable analysis, the highest risk for maternal GBS colonization was in mothers from Sub-Saharan (aRR = 1.45, 95%CI: 1.41–1.48) and North-African origin (aRR = 1.31, 95%CI: 1.28–1.34), those with parity of 5 or more (aRR = 1.21, 95%CI: 1.18–1.25), mothers with obesity (aRR = 1.09, 95%CI: 1.08–1.11), and pregnant women having received very little formal education (aRR = 1.09, 95%CI: 1.06–1.13).

Neonatal GBS infection

Between 2012 and 2021, 1,204,802 babies were born alive in Belgium, with an annual average of 120,480. Fifty-four percent was born in Flanders, 31% in Wallonia and 15% in the Brussels-Capital Region.

From 2016 to 2021, 543 cases of neonatal GBS sepsis and 5,580 unspecified cases of neonatal bacterial sepsis were recorded in the MHD. The incidence of both neonatal GBS sepsis and overall neonatal bacterial sepsis decreased in this period (from 1.26 to 0.29 cases per 1000 livebirths and from 8.78 to 6.46 cases per 1000 livebirths, respectively). The proportion of GBS neonatal sepsis in overall neonatal sepsis also gradually decreased Fig. 1. Overall, GBS neonatal sepsis incidence was 0.76 cases per 1000 livebirths, 0.88 for males and 0.64 for females.

Between 2012 and 2021, the BBR registered 625,836 neonates in Flanders (data from Brussels/Wallonia not available), of whom 622,672 were born alive. Of them, 305 were confirmed to have neonatal GBS sepsis. Seven

Table 1 GBS screening status of pregnant women giving live birth by year in Belgium, 2012–2021 ($n = 1,197,994$), data source: Belgian Birth Register

Year of birth	All women with live birth	Screened	Not screened	Missing	Not screened excl. CS or delivery < 35 weeks	Colonized
2012	126,737	90.3%	9.7%	0.0%	6.2%	17.4%
2013	123,961	90.3%	9.6%	0.0%	6.0%	16.9%
2014	123,558	91.0%	8.9%	0.1%	5.6%	17.3%
2015	120,879	91.0%	9.0%	0.0%	5.5%	17.3%
2016	120,532	90.6%	9.3%	0.1%	5.8%	16.9%
2017	118,329	90.5%	9.5%	0.0%	5.8%	17.1%
2018	117,436	90.0%	9.9%	0.1%	5.9%	16.7%
2019	116,591	90.1%	9.9%	0.0%	6.0%	16.1%
2020	113,042	88.6%	11.2%	0.3%	7.3%	15.5%
2021	116,929	89.9%	9.7%	0.3%	5.8%	15.4%
Total	1,197,994	90.2%	9.7%	0.1%	6.0%	16.7%

If screening status was not reported/unknown, data is shown as missing

Table 2 Proportion of GBS colonized and non-colonized pregnant women by risk factor, uni- and multivariable analysis of risk factors for maternal GBS colonization, 2012–2021, Belgium, data source: Belgian Birth Register

Variables	GBS-colonized	GBS non-colonized	Baseline risk	Univariable analysis RR [CI]	Multivariable analysis aRR [CI]
	<i>n</i> = 199,669 (100.0%)	<i>N</i> = 881,416 (100.0%)			
Maternal age					
< = 20 yrs	1.8%	1.9%		0.97 [0.95–1.00]**	1.01 [0.97–1.04]**
> 20–34 yrs	76.9%	78.4%	18%	1	
35 + yrs	21.3%	19.7%		1.08 [1.07–1.09]**	1.04 [1.03–1.06]**
Parity					
Primipara	42.0%	44.4%	17%	1	
Parity 2–4	54.4%	52.9%		1.06 [1.05–1.07]**	1.04 [1.03–1.05]**
Parity > = 5	3.6%	2.6%		1.30 [1.28–1.33]**	1.21 [1.18–1.25]**
Hypertension					
No	95.5%	95.7%	18%	1	
Yes	4.5%	4.3%		1.05 [1.03–1.07]**	1.04 [1.02–1.06]**
Diabetes					
No	92.1%	92.9%	18%	1	
Yes	7.9%	7.1%		1.08 [1.06–1.10]**	1.02 [1.00–1.04]*
BMI					
< = 18.5	4.3%	4.9%		0.94 [0.92–0.95]**	0.94 [0.92–0.96]**
18.5–25	56.7%	59.0%	17%	1	
25–30	24.3%	22.9%		1.07 [1.06–1.08]**	1.04 [1.03–1.05]**
> 30	14.5%	12.9%		1.13 [1.12–1.15]**	1.09 [1.08–1.11]**
Maternal education level					
No education/primary not completed	2.3%	1.6%		1.25 [1.21–1.28]**	1.09 [1.06–1.13]**
Primary/lower secondary	15.6%	14.0%		1.08 [1.07–1.09]**	1.01 [1.00–1.03]**
Upper secondary	31.0%	32.5%	17%	1	
More than secondary"uni"	45.9%	46.7%		1.03 [1.02–1.04]**	1.07 [1.06–1.08]**
Maternal work status					
Paid employment or student	66.8%	70.5%	17%	1	
No paid employment (housewife, unemployed)	23.3%	20.5%		1.12 [1.10–1.13]**	1.07 [1.05–1.08]**
Retired or incapacity to work	10.0%	9.0%		1.14 [1.12–1.16]**	1.07 [1.05–1.09]**
Maternal nationality					
Belgium	74.0%	77.9%	17%	1	
Northern/Southern/Western Europe	7.0%	7.4%		1.00 [0.98–1.01]**	0.98 [0.96–1.00]**
Eastern Europe/Russia	5.7%	5.3%		1.07 [1.05–1.08]**	1.07 [1.04–1.09]**
North Africa	5.3%	3.4%		1.37 [1.34–1.39]**	1.31 [1.28–1.34]**
Sub-Saharan Africa	4.7%	2.7%		1.51 [1.48–1.53]**	1.45 [1.41–1.48]**
Other	3.3%	3.3%		0.99 [0.96–1.01]**	0.96 [0.93–0.98]**

In multivariable analysis all variables were used

Confidence intervals are marked with * if corresponding *p*-values are < 0.05 and with ** if they are < 0.001

Baseline risk in the table refers to univariable models

In multivariable models baseline risk for GBS maternal colonization: 16%

babies were excluded from further analysis due to congenital malformations. For 186 included babies the reason for admission to neonatal ward was reported as

‘infection in the first seven days of life’, an additional 29 were reported to have a fetal infection and 58 a perinatal infection. Therefore, 92% (273/298) of included neonatal

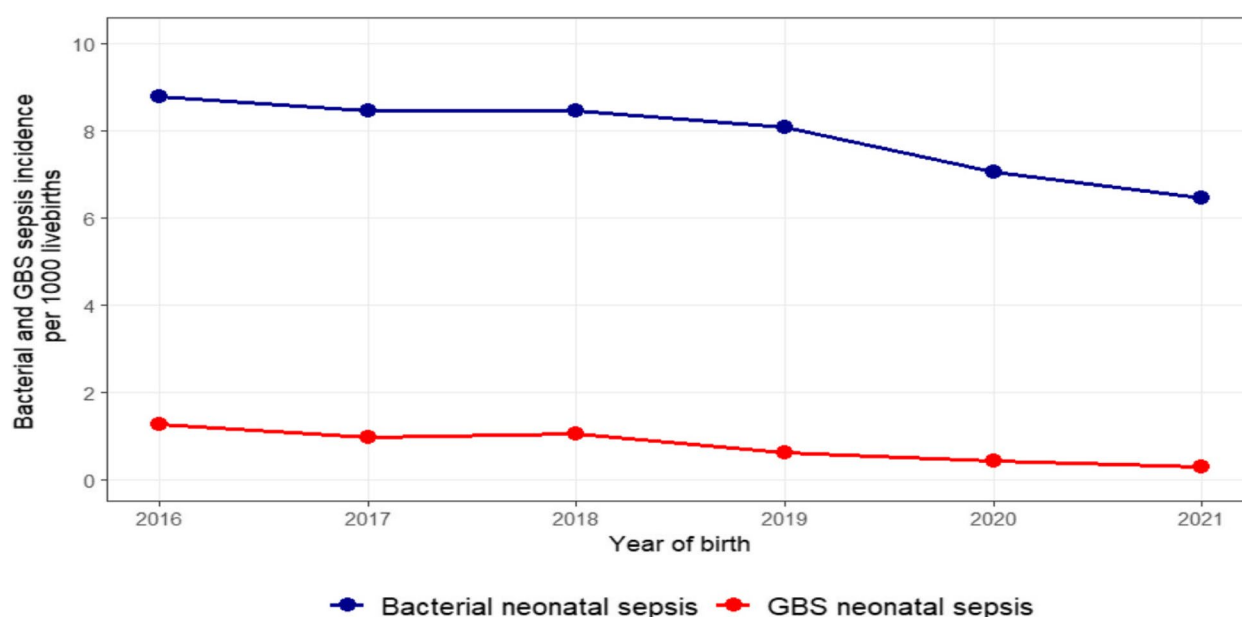


Fig. 1 Overall bacterial neonatal sepsis and GBS neonatal sepsis incidence (based on ICD-10 coding), Belgium, 2016–2021, ($N=5580$, $n=543$), data source: Minimal Hospital Data

GBS infections are indeed EOD cases. For 20 included cases, the infection was reported to be a nosocomial infection (> 72 h after birth) and these could either be LOD or EOD cases. For five cases, there was no information on the type of infection.

Based on these data, the overall reported incidence of neonatal GBS infections is 0.49 per 1000 livebirths. The annual incidence fluctuated over the study period and

showed a decreasing trend from 0.65 cases per 1000 livebirths in 2012 to 0.32 cases in 2021 (range: 0.32–0.70 cases per 1000 livebirths) Fig. 2. The neonatal incidence for male and female babies was 0.54 and 0.44 cases per 1000 livebirths, respectively. Of the neonatal GBS cases, 67% were born term. Of the 228 cases with mothers with known GBS status, 123 (54%) were born from GBS-positive mothers, of which 33 (27%) did not receive IAP.

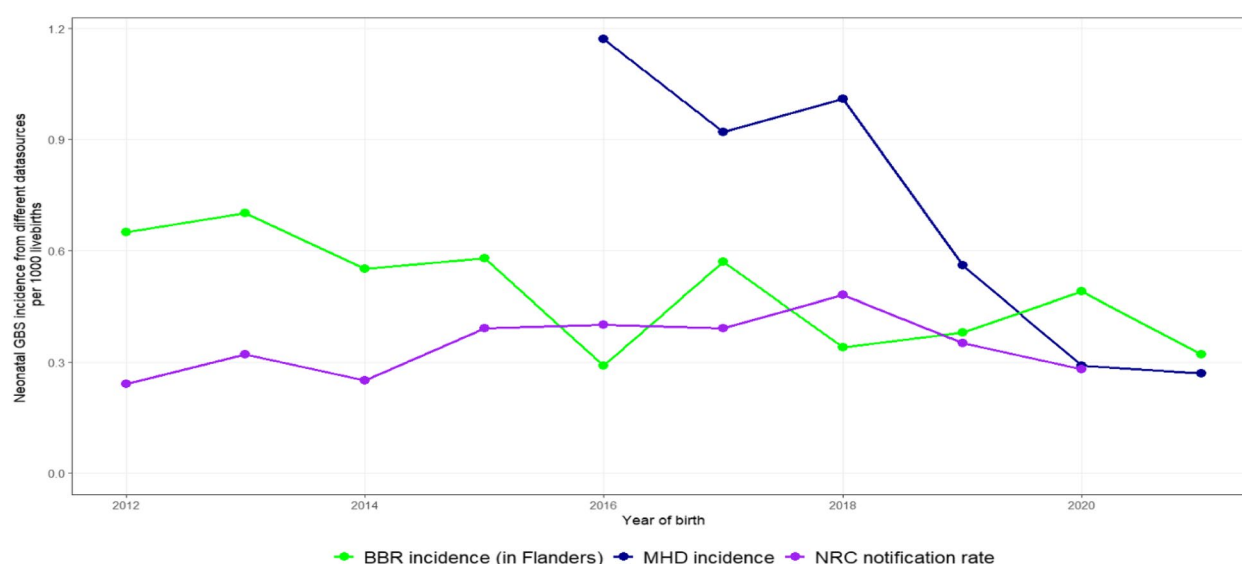


Fig. 2 Neonatal GBS incidences/notification rates from different data sources, Belgium, 2012–2020/21, data source: Minimal Hospital Data, National Reference Center for Group B *Streptococcus*, Belgian Birth Register in Flanders

Table 3 Case numbers, proportion of neonatal GBS infections by risk factor categories, uni- and multivariable analysis of risk factors for neonatal GBS infections, 2012–2021, Flanders, data source: Belgian Birth Register

Variables	Neonatal GBS infection <i>n</i> = 298 (100.0%)	Healthy babies <i>N</i> = 527 514 (100.0%)	Baseline risk	Univariable analysis RR [CI]	Multivariable analysis aRR [CI]
Baby's sex					
Female	166 (44.3%)	267,057 (49.4%)	0.05%	1	
Male	132 (55.7%)	260,455 (50.6%)		1.23 [0.98–1.54]	1.26 [1.01–1.58]*
Maternal GBS colonization + IAP					
Not screened	70 (23.5%)	46,876 (8.9%)		5.55 [4.10–7.50]**	1.65 [1.17–2.33]**
Not colonized	105 (35.2%)	390,498 (74.1%)	0.03%	1	
GBS colonized + no IAP	33 (11.1%)	13,323 (2.5%)		9.19 [6.22–13.59]**	7.045 [5.05–9.85]**
GBS colonized + IAP	90 (30.2%)	76,102 (14.4%)		4.39 [3.32–5.82]**	4.34 [3.28–5.74]**
Birth weight					
< 1000 g—extreme low birth weight	19 (6.4%)	362 (0.1%)		122 [77.13–192.98]**	3.25 [1.20–8.80]**
1001–1499 g—very low birth weight	22 (7.4%)	67 (0%)		604.72 [410.73–890.34]**	14.83 [6.48–33.94]**
1500–2499 g—low birth weight	41 (13.8%)	6096 (1.2%)		16.34 [11.71–22.81]**	6.72 [4.44–10.16]**
2500–4499 g	211 (70.8%)	515,976 (97.8%)	0.04%	1	
> 4500 g	5 (1.7%)	5013 (1%)		2.44 [1.00–5.91]**	2.49 [1.03–6.07]**
Maternal age					
< = 20	2 (0.7%)	8448 (1.6%)		0.43 [0.11–1.70]	0.36 [0.09–1.44]
> 20–34	234 (78.5%)	420,002 (79.6%)	0.06%	1	
35 +	62 (20.8%)	99,053 (18.8%)		1.12 [0.85–1.49]	1.02 [0.78–1.35]
Gestational age					
Extreme preterm < 28w	22 (7.4%)	358 (0.1%)		147.95 [96.40–227.07]**	40.83 [16.44–101.40]**
Very preterm from 28 to less than 32 w	26 (8.7%)	89 (0%)		577.76 [401.02–832.42]**	53.42 [23.65–120.66]**
Moderate preterm from 32 to less than 37 w	48 (16.1%)	11,050 (2.1%)		11.05 [8.07–15.13]**	7.06 [4.86–10.24]**
Term	202 (67.8%)	516,008 (97.8%)	0.04%	1	
Maternal nationality					
Belgium	222 (76.8%)	430,057 (82.9%)	0.05%	1	
Northern/Southern/Western Europe	17 (5.9%)	26,563 (5.1%)		1.24 [0.76–2.03]*	1.17 [0.72–1.91]*
Eastern Europe/Russia	23 (8%)	24,571 (4.7%)		1.81 [1.18–2.78]*	1.83 [1.20–2.80]*
North Africa	11 (3.8%)	11,155 (2.1%)		1.91 [1.04–3.50]*	1.73 [0.99–3.05]*
Sub-Saharan Africa	10 (3.5%)	8847 (1.7%)		2.19 [1.16–4.12]*	1.63 [0.89–3.02]*
Other	6 (2.1%)	17,715 (3.4%)		0.66 [0.29–1.48]*	0.66 [0.29–1.48]*

Missing cases are not displayed in the table

Confidence intervals are marked with * if corresponding *p*-values are < 0.05 and with ** if they are < 0.001

Baseline risk in the table refers to univariable models

In multivariable models baseline risk for neonatal GBS infection: 0.02%

Seventy-three percent had sepsis, 3.4% meningitis, 2.6% pneumonia and 3.6% died. Among GBS infected babies born preterm, 61% had not received IAP.

The NRC recorded 374 invasive neonatal cases between 2012 and 2020 in the whole of Belgium. Of these, 49% had EOD, 48% LOD and 3% VL0D. Of all neonatal infections, 53% were male and 47% were female. Clinical manifestation was missing for 30% of cases. Of the remaining,

68% had sepsis, 28% meningitis, 2% respiratory infection, 1% bone and joint infection or soft tissue and skin infection and less than 1% had urinary infection. Five cases (2% of all cases with known outcome) died. The annual notification rate ranged between 0.24 and 0.48 per 1000 livebirths. No invasive neonatal strains were resistant to penicillin or vancomycin, but 23% were resistant to clindamycin and 32% to erythromycin.

Table 3 presents the result of the neonatal GBS infection risk factor analysis. In multivariable analysis, the strongest risk factors were preterm birth (extreme, very and late preterm aRR = 40.83, 95%CI: 16.44–101.40; aRR = 53.42, 95%CI: 23.65–120.66; aRR = 7.06, 95%CI: 4.86–10.24, respectively) and very low and low birthweight (aRR = 14.83, 95%CI: 6.48–33.94; aRR = 6.72, 95%CI: 4.44–10.16, respectively). The risk ratio of neonatal GBS infection was higher among colonized mothers without IAP (aRR = 7.05, 95%CI: 5.05–9.85) than with IAP (aRR = 4.34, 95%CI: 3.28–5.74) when comparing to non-colonized women. Males had a higher risk to acquire neonatal GBS infection (aRR = 1.26, 95%CI: 1.01–1.58) compared to females (Table 3).

Discussion

We presented the first comprehensive overview of the epidemiological situation and risk factors of maternal GBS colonization and neonatal GBS disease in Belgium. We found that colonization rate was 18.5% among the Belgian pregnant population. At highest risk of maternal GBS colonization were women of Sub-Saharan and North African origin, those with more than five deliveries, those with obesity and those with low education. In two independent data sources, we found a decreasing trend for neonatal GBS infections over the study period, from 0.65 cases per 1000 livebirths in 2012 to 0.32 cases in 2021 in BBR Flanders. Factors associated with the highest risk for invasive neonatal GBS infection were premature delivery, low birthweight and maternal GBS colonization independent of the administration of IAP.

The proportion of GBS screened mothers remained stable over the years. The proportion of maternal GBS colonization (18.5%) was also stable over the years and comparable to the previously estimated European prevalence of 19.7% (17.3%–22.6%) [1]. We have found African origin to be associated with higher risk of maternal GBS colonization, which is in line with several other studies [30–32]. The reason behind racial disparities needs further investigation, however, it is already known that vaginal microbiomes vary significantly among women of different ethnicities [33]. We also found an increased risk with increasing BMI, which was supported by an American study [34]. As in our study, an Italian study found low education to be a risk factor for maternal colonization [31]. While mechanisms are to be further elucidated, affected immune system, access to health care and nutrition might all contribute to this increased risk for GBS colonization. Although we found a large number of risk factors for GBS colonization, an approach to identify mothers with GBS colonization based solely on risk factors proves challenging in clinical practice as most risk factors either have only a weak association or are indeed rare (e.g. parity 5+).

The Belgian guideline [20] indicates universal screening at 35–37 th week gestational age, so non-screened women include those with preterm delivery. Moreover, intrapartum prophylaxis is not indicated in case of a planned caesarean delivery (CS). Non-screening does hence not necessarily indicate non-compliance with guidelines. The proportion of non-screened pregnant women decreased substantially to 6% when premature deliveries and planned CS were excluded. This group includes both women requiring IAP by default i.e., those with GBS bacteriuria during the current pregnancy or a previous neonate with GBS infection, and women for whom the recommendations were not adhered to. There were more non-screened women during the pandemic year 2020, which might either be due to poor registration or difficulties in accessing healthcare. Moreover, 61% of preterm babies did not receive IAP. Whether this signals non-adherence to guidelines or circumstances that do not allow for IAP (e.g. sudden preterm delivery), we cannot determine.

We found similar incidence and notification rates of neonatal GBS infection in different Belgian data sources. Overall BBR estimates were around 0.49 cases per 1000 livebirths in Flanders, which is comparable to European estimates (0.53 per 1000 livebirths) [35]. We identified a decreasing neonatal GBS incidence trend in the MHD and BBR data. During the period when data were comparable from all data sources (2016–2021), a stable trend of neonatal GBS incidence was observed in BBR and NRC data. The widest range of incidence estimates was found in MHD, with the highest incidence coinciding with the launch and the rollout of the new ICD-10 code. Although the reason is unknown, biased coding practices cannot be excluded. While we observed a stable trend in the other 2 datasources, in MHD both overall bacterial sepsis and GBS sepsis incidence showed a decreasing trend. Compared to unspecified overall bacterial neonatal sepsis, only few cases are coded as neonatal GBS sepsis. Moreover, due to the data structure and selection process, included neonatal GBS infections from the BBR data were 92% EOD cases. In contrast, NRC data showed 48% of the total neonatal GBS cases as LOD. It is therefore probable that our results are an underestimation of the true GBS case numbers. The incidence of invasive GBS infection was higher among male babies in all datasets, in line with earlier literature findings [36]. Differences in infectious morbidity and mortality between male and female neonates is known to be related to sex hormones' effect on the immune system [37]. Importantly, for almost half of the neonatal GBS cases with known maternal screening results in BBR data, maternal colonization was reported negative. This may be due to transient colonization, inadequate sampling or issues with culturing despite of guideline recommendation of using specific, GBS selective media and represents

an issue for the universal screening as a stand-alone strategy. Indeed, previous studies have reported similar issues [1, 38]. Among colonized women, the risk of neonatal GBS infection was higher among those without IAP than in those who received IAP, indicating the effectiveness of IAP in preventing EOD cases. In contrast, in absolute numbers, 90 neonatal GBS cases were registered from GBS-positive mothers that had indeed received IAP, compared to 33 from GBS-positive mothers who did not receive the IAP. Additionally, more than 1 in four (27%) of women with known GBS-colonization did not receive IAP. In this study, we could not determine reasons for not administering IAP. Potential reasons include both logistic issues (e.g. in case of precipitous labor) as well as non-awareness or non-adherence to guidelines. An additional weakness of the IAP prevention strategy is that it does not affect LOD. NRC data included in our study, as well as a European multistate study [39] and an American study [13] estimated that about half of all neonatal iGBS cases are indeed LOD.

Although we did not find any resistance to the first-line treatment penicillin, almost 25% of the strains were resistant to clindamycin, with an increasing trend. Clindamycin is recommended as IAP in case of penicillin allergy (after resistance testing). Recently, WHO has been highlighting the potential of vaccinations in general as a tool to reduce antimicrobial resistance [40].

Babies born very and extreme premature had the highest risk to acquire neonatal GBS infection, as has also been documented elsewhere in international literature [28, 38]. We find indeed that the risk for very preterm infants is more than 570 times that of babies born at term. Since preterm babies importantly overlap with low-birthweight babies, the adjusted relative risk for very preterm infants reduces to 53 in the multivariable analysis because part of the association is captured by birthweight. To protect those mothers and their babies, a future vaccine should ideally achieve protection early in pregnancy. IgG transfer through the placenta starts actively from gestational week 28 [41] and reaches its maximum by the end of the pregnancy. We found only 0.1% of GBS infected neonates were born before 28 weeks, indicating the usefulness of vaccination preventing neonatal GBS infections. Maternal infection during pregnancy increases the risk of premature birth, so GBS vaccines may also reduce preterm birth itself [42, 43]. GBS vaccines have the potential to also protect against LOD. This is important, as NRC data estimated 48% of cases being LOD. Additionally, vaccination could also decrease the amount of administered IAP with possible effects on the neonatal microbiome as have been shown in studies [44, 45].

Limitations and strengths

Our study has limitations. Because of concerns about reidentification, a small cell policy was used for MHD

data, causing us to use a narrow definition of invasive neonatal GBS disease as neonatal GBS sepsis. Underestimation of neonatal cases is therefore expected, and our reported incidences should be interpreted as minima. Changes in the coding system in MHD during the study period resulted in serious data quality differences, so only the period with ICD-10 data was used for neonatal GBS infections. Neither the data from MHD or BBR allowed a formal distinction between EOD and LOD. However, the strength of our analysis lies in the use of multiple data sources to cross-validate results and a very large dataset on maternal colonization. Nationwide data on GBS colonization and prophylaxis use are rare in other studies.

Conclusion

These baseline data provide insights in the current Belgian situation and could help tailor additional future preventive strategies such as maternal vaccination. Existing recommendations of IAP should be reinforced and could in the future be complemented by a vaccination strategy to further reduce neonatal GBS morbidity, antibiotic use and prevent antimicrobial resistance.

Although several data sources are available, delays are long, data quality is varied and used definitions are inconsistent. Data availability should be optimized to allow timely follow-up of the epidemiological situation, especially in case of changes to the prevention strategy.

Abbreviations

BBR	Belgian Birth Register
CEpiP	Centre d'Épidémiologie Périnatale, Perinatal Epidemiology Center
CS	Cesarean section
EOD	Early onset disease
IAP	Intrapartum antibiotic prophylaxis
ICD	International Classification of Diseases
iGBS	Invasive group B Streptococcus
LOD	Late onset disease
MHD	Minimal Hospital Data
NRC	National Reference Center for Group B Streptococcus
SPE	Study Center for Perinatal Epidemiology
VLOD	Very late onset of disease
WHO	World Health Organization

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

Study conception and design ZsB, LC; data collection ZsB, LC; data cleaning and analysis ZsB, FDK; all authors contributed to the interpretation of results, original draft manuscript preparation ZsB; manuscript revision LC, FDK, AL, CL, VWL, RG, PM, CM, RS, FC.

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Data availability

The data that support the findings of this study are available from the primary data owners but restrictions apply to the availability of these data, which were used under license for the current study and are as such not publicly available. Data are, however, available from the authors upon reasonable request and conditional on obtaining permission from the different data owners. Data owners can be contacted at regine.goemaes@vlaanderen.be (SPE data), perinatalite@cepip.be (CEPIP data), info.rhmzg@gezondheid.belgie.be (MHD) or nrcm@sciensano.be (NRC data).

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. All data accessed in the context of the present study had been previously collected as part of the routine data collection for epidemiological surveillance and national statistics. In accordance with article 9 of the Royal Decree of 14/07/1999 on yearly birth statistics, these data can be used for further research purposes. Therefore, our study was not submitted to a local ethics committee or institutional review board (IRB) because the need for ethical approval was deemed unnecessary according to national regulations. Following article 6, §1, (e) of the General Data Protection Regulation, no written informed consent from the patients is required for the collection and analysis of epidemiological data when the processing of data is necessary for the performance of a task carried out in the public (health) interest. No additional data was collected for the sole purpose of this study, and we used only non-identifiable pseudonymized data. Therefore, the need for consent to participate in this study was deemed unnecessary.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

- Puopolo KM, Lynfield R, Cummings JJ, Committee on fetus and newborn, Committee on infectious diseases, Hand I, et al. Management of infants at risk for group B streptococcal disease. *Pediatrics*. 2019;144(2):e20191881.
- RCOG. Prevention of early-onset group B streptococcal disease (Green-top Guideline No. 36). 2024. Available from: <https://www.rcog.org.uk/guidance/browse-all-guidance/green-top-guidelines/prevention-of-early-onset-group-b-streptococcal-disease-green-top-guideline-no-36/>. Cited 2024 Aug 22.
- Prevention of group B streptococcal early-onset disease in newborns. 2024. Available from: <https://www.acog.org/clinical/clinical-guidance/committee-opinion/articles/2020/02/prevention-of-group-b-streptococcal-early-onset-disease-in-newborns>. Cited 2024 Mar 7.
- Gonçalves BP, Procter SR, Paul P, Chandna J, Lewin A, Seedat F, et al. Group B streptococcus infection during pregnancy and infancy: estimates of regional and global burden. *Lancet Glob Health*. 2022;10(6):e807–19.
- Task force for group B streptococcal infections for the Italian Society of Neonatology, Italian Society of Obstetricians and Gynecologists, Italian Association of Clinical Microbiologists, Tzialla C, Berardi A, Farina C, et al. Strategies for preventing group B streptococcal infections in newborns: a nation-wide survey of Italian policies. *Ital J Pediatr*. 2017;43(1):98.
- Rosa-Fraile M, Alós JI. Group B streptococcus neonatal infections, the ongoing history. *Enferm Infecc Microbiol Clin*. 2022;40(7):349–52.
- Klemetti R, Hakulinen-Viitanen T. Äitiysneuvolaopas - Suosituksia äitiysneuvolatoimintaan. THL; 2013. Available from: <https://www.julkari.fi/handle/10024/110521>. Cited 2024 Apr 19.
- Morita JY, O'Brien KL, Schuchat A. Prevention of perinatal group B streptococcal infections. *Pediatr Infect Dis J*. 1999;18(3):279–80.
- Royal College of Obstetricians and Gynaecologists (RCOG) Green-top Guideline. Prevention of early-onset neonatal group B streptococcal disease. *BJOG: Int J Obstet Gynaecol*. 2017;124(12):e280–305.
- Protocol and documents. Available from: <https://www.gbs3trial.ac.uk/health-care-professionals/protocol-and-documents.aspx>. Cited 2024 Dec 17.
- Intrapartum antibiotica op basis van risicofactoren bij early-onset neonatale infecties - Richtlijn - Richtlijndatabase. 2024. Available from: https://richtlijndatabase.nl/richtlijn/preventie_en_behandeling_van_early-onset_neonatale_infecties/intrapartum_antibiotica_op_basis_van_risicofactoren_bij_early-onset_neonatale_infecties.html. Cited 2024 Aug 22.
- Björklund V, Saxén H, Hertting O, Carlsen ELM, Hoffmann S, Håkansson S, et al. Early-onset group B streptococcal infections in five Nordic countries with different prevention policies, 1995 to 2019. *Eurosurveillance*. 2024;29(3):2300193.
- Nanduri SA, Petit S, Smelser C, Apostol M, Alden NB, Harrison LH, et al. Epidemiology of invasive early-onset and late-onset group B streptococcal disease in the United States, 2006 to 2015. *JAMA Pediatr*. 2019;173(3):224–33.
- Hansen SM, Uldbjerg N, Kilian M, Sørensen UBS. Dynamics of streptococcus agalactiae colonization in women during and after pregnancy and in their infants. *J Clin Microbiol*. 2004;42(1):83–9.
- Morreale C, Giaroni C, Baj A, Folgori L, Barcellini L, Dhami A, et al. Effects of perinatal antibiotic exposure and neonatal gut microbiota. *Antibiotics (Basel)*. 2023;12(2):258.
- Doare K, O'Driscoll M, Turner K, Seedat F, Russell NJ, Seale AC, et al. Intrapartum antibiotic chemoprophylaxis policies for the prevention of group B streptococcal disease worldwide: systematic review. *Clin Infect Dis*. 2017;65(Suppl 2):S143–51.
- Prevention of group B streptococcal early-onset disease in newborns. Available from: <https://www.acog.org/clinical/clinical-guidance/committee-opinion/articles/2020/02/prevention-of-group-b-streptococcal-early-onset-disease-in-newborns>. Cited 2024 Mar 7.
- Group B Streptococcus (GBS). Available from: [https://www.who.int/teams/immunization-vaccines-and-biologicals/diseases/group-b-streptococcus-\(gbs\)](https://www.who.int/teams/immunization-vaccines-and-biologicals/diseases/group-b-streptococcus-(gbs)). Cited 2023 Jul 6.
- Defeating meningitis by 2030: a global road map. Available from: <https://www.who.int/publications-detail-redirect/9789240026407>. Cited 2023 Jul 6.
- Prévention des infections périnatales à streptocoques du groupe B (avril 2003) (CSH 7721).pdf. Available from: https://www.health.belgium.be/sites/default/files/uploads/fields/fpshealth_theme_file/4448391/Pr%C3%A9vention%20des%20infections%20p%C3%A9rinatales%20C3%A0%20streptocoques%20du%20groupe%20B%20%28avril%202003%29%20%28CSH%207721%29.pdf. Cited 2023 Jul 6.
- Minimum Hospital Data Set | European Health Information Portal. Available from: <https://www.healthinformationportal.eu/health-information-sources/minimum-hospital-data-set>. Cited 2024 Feb 23.
- Khan MA, Faiz A, Ashshi AM. Maternal colonization of group B streptococcus: prevalence, associated factors and antimicrobial resistance. *Ann Saudi Med*. 2015;35(6):423–7.
- Maternal age and body mass index as risk factors for rectovaginal colonization with group B streptococci - Khalil - 2023 - International Journal of Gynecology & Obstetrics - Wiley Online Library. Available from: <https://obgyn.onlinelibrary.wiley.com/doi/full/10.1002/ijgo.14449>. Cited 2024 Jul 11.
- Gopal Rao G, Hiles S, Bassett P, Lamagni T. Differential rates of group B streptococcus (GBS) colonisation in pregnant women in a racially diverse area of London, UK: a cross-sectional study. *BJOG*. 2019;126(11):1347–53.
- Capan-Melser M, Mombo Ngoma G, Akerey-Diop D, Basra A, Würbel H, Groger M, et al. Evaluation of intermittent preventive treatment of malaria against group B Streptococcus colonization in pregnant

women: a nested analysis of a randomized controlled clinical trial of sulfadoxine/pyrimethamine versus mefloquine. *J Antimicrob Chemother.* 2015;70(6):1898–902.

26. Hung LC, Kung PT, Chiu TH, Su HP, Ho M, Kao HF, et al. Risk factors for neonatal early-onset group B streptococcus-related diseases after the implementation of a universal screening program in Taiwan. *BMC Public Health.* 2018;18(1):438.
27. Mukhopadhyay S, Puopolo KM. Risk assessment in neonatal early onset sepsis. *Semin Perinatol.* 2012;36(6):408–15.
28. Adair CE, Kowalsky L, Quon H, Ma D, Stoffman J, McGeer A, et al. Risk factors for early-onset group B streptococcal disease in neonates: a population-based case–control study. *CMAJ.* 2003;169(3):198–203.
29. Schuchat A, Oxtoby M, Cochi S, Sikes RK, Hightower A, Plikaytis B, et al. Population-based risk factors for neonatal group B streptococcal disease: results of a cohort study in metropolitan Atlanta. *J Infect Dis.* 1990;162(3):672–7.
30. Gopal Rao G, Hiles S, Bassett P, Lamagni T. Differential rates of group B streptococcus (GBS) colonisation in pregnant women in a racially diverse area of London, UK: a cross-sectional study. *BJOG: Int J Obstetr Gynaecol.* 2019;126(11):1347–53.
31. Valkenburg-van den Berg AW, Sprij AJ, Oostvogel PM, Mutsaers JAEM, Renes WB, Rosendaal FR, et al. Prevalence of colonisation with group B Streptococci in pregnant women of a multi-ethnic population in The Netherlands. *Eur J Obstet Gynecol Reprod Biol.* 2006;124(2):178–83.
32. Kwatra G, Cunningham MC, Merrill E, Adrian PV, Ip M, Klugman KP, et al. Prevalence of maternal colonisation with group B streptococcus: a systematic review and meta-analysis. *Lancet Infect Dis.* 2016;16(9):1076–84.
33. Ravel J, Gajer P, Abdo Z, Schneider GM, Koenig SSK, McCulle SL, et al. Vaginal microbiome of reproductive-age women. *Proc Natl Acad Sci U S A.* 2011;108 Suppl 1(Suppl 1):4680–7.
34. Venkatesh KK, Vladutiu CJ, Strauss RA, Thorp JM, Stringer JSA, Stamilio DM, et al. Association between maternal obesity and group B streptococcus colonization in a National U.S. Cohort. *J Womens Health (Larchmt).* 2020;29(12):1507–12.
35. Madrid L, Seale AC, Kohli-Lynch M, Edmond KM, Lawn JE, Heath PT, et al. Infant group B streptococcal disease incidence and serotypes worldwide: systematic review and meta-analyses. *Clin Infect Dis.* 2017;65(suppl_2):S160–72.
36. van Kassel MN, Gonçalves BP, Snoek L, Sørensen HT, Bijlsma MW, Lawn JE, et al. Sex differences in long-term outcomes after group B streptococcal infections during infancy in Denmark and the Netherlands: National Cohort Studies of Neurodevelopmental Impairments and Mortality. *Clin Infect Dis.* 2022;74(Suppl_1):S54–63.
37. Muenchhoff M, Goulder PJR. Sex differences in pediatric infectious diseases. *J Infect Dis.* 2014;209(Suppl 3):S120–6.
38. Berardi A, Trevisani V, Di Caprio A, Bua J, China M, Perrone B, et al. Understanding factors in group B streptococcus late-onset disease. *Infect Drug Resist.* 2021;17(14):3207–18.
39. Lohrmann F, Hufnagel M, Kunze M, Afshar B, Creti R, Detscheva A, et al. Neonatal invasive disease caused by *Streptococcus agalactiae* in Europe: the DEVANI multi-center study. *Infection.* 2023;51(4):981–91.
40. Leveraging vaccines to reduce antibiotic use and prevent antimicrobial resistance. Available from: <https://www.who.int/publications/m/item/leveraging-vaccines-to-reduce-antibiotic-use-and-prevent-antimicrobial-resistance>. Cited 2024 Aug 22.
41. Palmeira P, Quinello C, Silveira-Lessa AL, Zago CA, Carneiro-Sampaio M. IgG placental transfer in healthy and pathological pregnancies. *Clin Dev Immunol.* 2012;2012:985646.
42. Bianchi-Jassir F, Seale AC, Kohli-Lynch M, Lawn JE, Baker CJ, Bartlett L, et al. Preterm birth associated with group B streptococcus maternal colonization worldwide: systematic review and meta-analyses. *Clin Infect Dis.* 2017;65(Suppl 2):S133–42.
43. Absalon J, Simon R, Radley D, Giardina PC, Koury K, Jansen KU, et al. Advances towards licensure of a maternal vaccine for the prevention of invasive group B streptococcus disease in infants: a discussion of different approaches. *Hum Vaccin Immunother.* 2022;18(1):2037350.
44. García VR. Impact of intrapartum antibiotic prophylaxis for group B streptococcus on the term infant gut microbiome: a state of the science review. *J Midwifery Women's Health.* 2021;66(3):351–9.
45. Zou ZH, Liu D, Li HD, Zhu DP, He Y, Hou T, et al. Prenatal and postnatal antibiotic exposure influences the gut microbiota of preterm infants in neonatal intensive care units. *Ann Clin Microbiol Antimicrob.* 2018;17(1):9.

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